Disentangled Cost Learning for Air-Traffic Routing Under Potential-Shaping Invariance

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Abstract

We present a principled framework for learning edge-cost models in directed air-traffic networks while explicitly handling two sources of ambiguity: (1) the attribution ambiguity between feature-based costs and a residual preference, and (2) the potential-shaping invariance of routing policies. By projecting both features and residuals onto the cycle space of the graph, we obtain a gauge-fixed formulation

$$c = X_{\text{cyc}}\, w + p, \qquad BW\, p = 0,\ X_{\text{cyc}}^{\top} D p = 0,$$

which yields identifiable, policy-relevant cost components. The paper details the mathematical derivation, implementation recipes, and practical considerations.

1 Setup and notation

1.1 Graph and costs

- Directed graph $G = (V, E)$ with $|V| = n$ nodes and $|E| = m$ edges.
- Edge costs $c \in \mathbb{R}^m$, one scalar per edge.
- Edge-feature matrix $X \in \mathbb{R}^{m \times d}$, with d features per edge.
- Feature weights $w \in \mathbb{R}^d$.
- Residual / preference vector $p \in \mathbb{R}^m$.

1.2 Incidence matrix and potentials

Choose an orientation consistent with the directed edges and define the *full* node–edge incidence matrix $B_{\text{full}} \in \mathbb{R}^{n \times m}$ by

$$(B_{\text{full}})_{u,e} = \begin{cases} +1 & \text{if } e \text{ leaves node } u, \\ -1 & \text{if } e \text{ enters node } u, \\ 0 & \text{otherwise.} \end{cases}$$

A node potential $\phi \in \mathbb{R}^n$ induces an edge "gradient"

$$B_{\text{full}}^\top \phi \in \mathbb{R}^m, \qquad \left(B_{\text{full}}^\top \phi\right)_e = \phi(u) - \phi(v)$$

for $e : u \to v$.

To remove the constant-potential nullspace we anchor one node (typically a goal) by dropping its row, obtaining the *reduced* incidence matrix

$$B \in \mathbb{R}^{(n-1)\times m}.$$

Equivalently, we fix $\phi(\text{anchor}) = 0$ and represent ϕ only on the remaining $n-1$ nodes.

1.3 Weight matrices

We employ two diagonal edge-weight matrices:

- $W \in \mathbb{R}^{m\times m}$, $W \succ 0$, to define the gauge-fixing / cycle-space notion.
- $D \in \mathbb{R}^{m\times m}$, $D \succeq 0$, to define the disentanglement notion.

Typical choices are $W = I$ (global gauge fix) or $W = D_\epsilon = \operatorname{diag}(\hat{n} + \epsilon)$ with a small $\epsilon > 0$, and an analogous choice for D.

2 Why costs are not identifiable: potential-shaping invariance

2.1 Deterministic shortest path

For any origin–goal pair $o \to g$ and a path $\tau = (e_1, \ldots, e_T)$, the path cost is $\sum_{t=1}^{T} c_{e_t}$. If we add a potential gradient to edge costs,

$$c' = c + B_{\text{full}}^\top \phi,$$

then

$$\begin{aligned} \sum_{t=1}^{T} c'_{e_t} &= \sum_{t=1}^{T} c_{e_t} + \sum_{t=1}^{T} \big(\phi(s_t) - \phi(s_{t+1})\big) \\ &= \sum_{t=1}^{T} c_{e_t} + \phi(o) - \phi(g). \end{aligned} \tag{1}$$

The shift $\phi(o) - \phi(g)$ is the same for all $o \to g$ paths, so the argmin path(s) are unchanged.

2.2 Maximum-entropy (soft) path distributions

In a MaxEnt model,

$$P(\tau \mid o, g, c) = \frac{\exp\big(-c^\top n(\tau)\big)}{Z_{o,g}(c)}, \qquad n(\tau) \in \mathbb{R}^m \text{ counts edge usages.}$$

Adding the same potential gradient yields

$$(c + B_{\text{full}}^\top \phi)^\top n(\tau) = c^\top n(\tau) + \phi(o) - \phi(g),$$

so the constant cancels between numerator and partition function $Z_{o,g}$; the distribution is unchanged.

2.3 Consequence

The policy (or likelihood) identifies only an equivalence class

$$c \sim c + B_{\text{full}}^\top \phi.$$

Thus, writing $c = Xw + p$ without a gauge-fix allows the residual p to accidentally absorb an arbitrary $B^\top \phi$ component.

3 Two distinct ambiguities ("gauges") we must fix

3.1 Gauge A: feature–residual attribution ambiguity

Even with a fixed c, the decomposition $c = Xw + p$ is not unique unless we constrain p. Our proposal

$$X^\top D p = 0 \tag{1}$$

forces p to be orthogonal to the feature span under the D-weighted inner product. In words, "p contains no component explainable by the features (on the weighted support)."

3.2 Gauge B: potential-shaping ambiguity (policy invariance)

Even after fixing the split, c itself is ambiguous up to $B^\top \phi$. To ensure that p is a *policy-identifiable* preference we forbid p from containing any gradient component. This is expressed by the *cycle-space constraint*

$$BW\, p = 0. \tag{2}$$

Equivalently, Wp is a circulation (zero divergence at every node).

3.3 Combined requirement

Collecting (1) and (2) we require

$$p \in \ker\big(X^\top D\big) \;\cap\; \ker\big(BW\big). \tag{3}$$

4 Why $BWp = 0$ is the right "no potential component" condition

Define a W-weighted inner product on edge vectors:

$$\langle a, b \rangle_W := a^\top W b.$$

A vector p has no gradient component (in the W-orthogonal sense) iff it is orthogonal to every gradient $B^\top \phi$:

$$\langle p, B^\top \phi \rangle_W = 0 \quad \forall\, \phi.$$

Indeed,

$$\langle p, B^\top \phi \rangle_W = p^\top W B^\top \phi = (BWp)^\top \phi,$$

which vanishes for all ϕ exactly when $BWp = 0$. Hence (2) is precisely the desired condition.

5 The cycle projector P_{cyc} ("remove potential component")

5.1 Derivation via weighted least squares

Given any edge vector $v \in \mathbb{R}^m$ we seek the best-fitting potential gradient in the W-metric:

$$\phi^* = \arg\min_{\phi} \left\| v - B^\top \phi \right\|_W^2, \qquad \|x\|_W^2 := x^\top W x.$$

The first-order optimality condition reads

$$B\,W\big(v - B^\top \phi^*\big) = 0 \iff (BWB^\top)\phi^* = BWv.$$

Let the reduced weighted Laplacian be

$$L := BWB^\top \in \mathbb{R}^{(n-1)\times(n-1)}.$$

Assuming $W \succ 0$ and a proper anchor, L is symmetric positive definite. Thus

$$\phi^* = L^{-1}(BWv), \qquad P_{\text{cyc}} v := v - B^\top \phi^*. \tag{4}$$

Equation (4) can be written as a matrix operator (without forming it explicitly):

$$P_{\text{cyc}} = I - B^\top (BWB^\top)^{-1} BW. \tag{5}$$

5.2 Key properties

(i) **Gauge removal:**

$$P_{\text{cyc}}(v + B^\top \phi) = P_{\text{cyc}} v.$$

(ii) **Cycle constraint:**

$$BW\big(P_{\text{cyc}} v\big) = 0.$$

6 Project features into the cycle space once and for all

6.1 Why we must also project X

Even if p satisfies (2), the feature term Xw can hide shaping: if there exist Δw and ψ such that

$$X\Delta w = B^{\top}\psi,$$

then replacing $w \leftarrow w + \Delta w$ changes c only by a gradient, leaving the policy unchanged. Hence w is not identifiable.

6.2 Cycle-projected features

Define the *cycle-projected* feature matrix

$$X_{\text{cyc}} := P_{\text{cyc}}X. \tag{6}$$

By construction $BWX_{\text{cyc}} = 0$, so every column of X_{cyc} already lies in the cycle space.

6.3 Eliminating the shaping ambiguity in w

Suppose $X_{\text{cyc}}\Delta w = B^{\top}\psi$. Applying BW to both sides yields

$$0 = BWX_{\text{cyc}}\Delta w = L\psi,$$

and because L is invertible on the anchored component we obtain $\psi = 0$, hence $X_{\text{cyc}}\Delta w = 0$. Therefore, after (6) the only remaining non-identifiability of w is the usual linear-model nullspace (collinearity).

7 The recommended model

Putting the pieces together we adopt the gauge-fixed formulation

$$c = X_{\text{cyc}}\, w + p, \qquad \underbrace{BWp = 0}_{\text{no potential component}}, \qquad \underbrace{X_{\text{cyc}}^{\top}Dp = 0}_{\text{feature disentanglement}}. \tag{7}$$

Equation (7) yields a cost vector whose residual part p is

- orthogonal to the engineered features (weighted by D), and
- free of any gradient that could be absorbed by a potential-shaping transformation.

Consequently p captures a *policy-identifiable preference* over cycles, while $X_{\text{cyc}}w$ accounts for the known feature-based effects.

8 Practical end-to-end recipe

8.1 Precomputation

1. **Inputs**

   - Graph $G = (V, E)$.
   - Feature matrix $X \in \mathbb{R}^{m \times d}$.
   - Anchor node (often a goal).
   - Diagonal weight matrices $W \succ 0$, $D \succeq 0$.

2. **Build reduced incidence** B by dropping the anchor row.

3. **Form the weighted Laplacian**

$$L := B\,W\,B^{\top}.$$

   Factorize L (e.g., Cholesky) for repeated solves.

4. **Cycle-project the features**

   (a) Compute $R := BWX \in \mathbb{R}^{(n-1)\times d}$.

   (b) Solve $L\Phi_X = R$ for $\Phi_X \in \mathbb{R}^{(n-1)\times d}$.

   (c) Set $X_{\text{cyc}} := X - B^{\top}\Phi_X$.

5. **Pre-compute the D-weighted Gram matrix**

$$G := X_{\text{cyc}}^{\top} D X_{\text{cyc}} + \gamma I,$$

   with a small ridge $\gamma > 0$; factorize G as well.

8.2 Training loop (projected gradient)

The following pseudocode assumes we can obtain the gradient of the MaxEnt log-likelihood with respect to the edge costs c (denoted g_c).

```
# Precomputed objects:
#   B, W, D
#   L = B @ W @ B.T   (factorized)
#   X_cyc
#   G = X_cyc.T @ D @ X_cyc + gamma * np.eye(d)   (factorized)

# hyperparameters
eta_w = 1e-3
eta_p = 1e-3
lambda_w = 1e-4
lambda_p = 1e-4

# initialization
```

```python
w = np.zeros(d)
p = np.zeros(m)   # already satisfies BWp=0 and X_cyc.T @ D @ p = 0

for t in range(T):
    # 1) current costs
    c = X_cyc @ w + p

    # 2) obtain model expected edge counts under current costs
    #    (implementation depends on the IRL algorithm)
    n_model = expected_edge_counts(c, demonstrations)

    # 3) gradient of negative log likelihood w.r.t. c
    g_c = n_hat - n_model   # empirical minus expected

    # 4) gradient step for w
    w -= eta_w * (X_cyc.T @ g_c + lambda_w * (w - w0))

    # 5) tentative update of p
    p_tmp = p - eta_p * (g_c + lambda_p * p)

    # 6) --- project p onto the intersection ---
    # (a) cycle projection: p1 = P_cyc(p_tmp)
    rhs = B @ (W @ p_tmp)
    phi = solve(L, rhs)              # L^{-1} rhs
    p1 = p_tmp - B.T @ phi           # now BW p1 = 0 (up to numerical error)

    # (b) D weighted feature orthogonalisation:
    b = X_cyc.T @ (D @ p1)
    alpha = solve(G, b)              # G^{-1} b
    p = p1 - X_cyc @ alpha           # preserves BW p = 0 and enforces X_cyc^T D p = 0
```

Listing 1: Projected gradient for w and p

After convergence we return (w, p) and the final cost $c = X_{\text{cyc}} w + p$.

8.3 Why the two-step projection is exact

Because $BWX_{\text{cyc}} = 0$, any vector in the column span of X_{cyc} already satisfies the cycle constraint. Hence after step (a) we have $BWp_1 = 0$, and subtracting $X_{\text{cyc}}\alpha$ in step (b) leaves the constraint unchanged:

$$BW(p_1 - X_{\text{cyc}}\alpha) = BWp_1 - BWX_{\text{cyc}}\alpha = 0.$$

9 What if we do *not* pre-project X?

If X is used directly, the two-step projection may re-introduce a gradient component when we subtract $X\alpha$ (because BWX need not be zero). In that case we must project onto the intersection $\ker(X^\top D) \cap \ker(BW)$ in a single solve:

$$\mathrm{Proj}(v) = v - A^\top \lambda, \qquad A = \begin{bmatrix} X^\top D \\ BW \end{bmatrix}, \ (AA^\top)\lambda = Av.$$

While mathematically correct, this approach is computationally heavier than the pre-projected strategy described above.

10 Interpretation of each constraint

Component	Constraint	Meaning
Disentanglement	$X_{\mathrm{cyc}}^\top D p = 0$	p contains no component explainable by the engineered features (v
Gauge-fix (residual)	$BWp = 0$	p contains no potential-shaping (gradient) component.
Feature gauge-fix	$X_{\mathrm{cyc}} = P_{\mathrm{cyc}} X$	Features themselves are already free of gradient components.

Table 1: Summary of the three core constraints.

11 Remaining identifiability caveats

1. **Feature collinearity.** If $X_{\mathrm{cyc}} \Delta w = 0$, w is not unique. Regularisation (ridge, sparsity) or explicit constraints can mitigate this.

2. **Pure-gradient features.** Any column of X that is exactly a gradient $B^\top \phi$ vanishes after projection (P_{cyc}), indicating that the feature is intrinsically unidentifiable from routing choices.

3. **Disconnected subgraphs.** If some nodes are unreachable from the anchor, the reduced Laplacian L is singular on those components. One must either restrict attention to the reachable subgraph or anchor each component separately.

4. **Scaling / temperature.** In MaxEnt models, multiplying c by a constant changes the effective temperature; this is independent of the potential-shaping gauge and must be handled (e.g., by fixing a temperature hyper-parameter).

12 Recommended defaults (practical)

- Choose $D = \mathrm{diag}(\hat{n} + \epsilon_D)$ and $W = \mathrm{diag}(\hat{n} + \epsilon_W)$ with a small $\epsilon > 0$ to avoid division by zero.

- Anchor a single, globally relevant node (e.g., a common destination).
- Add ℓ_2 regularisation on w and p, possibly weighted by D.
- Use a modest ridge γ when forming G to guarantee numerical stability.

13 Summary checklist

1. Build reduced incidence B and select a positive-definite W.
2. Factorize $L = BWB^\top$.
3. Compute the cycle-projected features $X_{\text{cyc}} = X - B^\top(BWB^\top)^{-1}BWX$.
4. Train the model $c = X_{\text{cyc}}w + p$ using projected gradient.
5. After each p update enforce:
   (i) $p \leftarrow P_{\text{cyc}}p$ (cycle/gauge projection);
   (ii) $p \leftarrow p - X_{\text{cyc}}\big(X_{\text{cyc}}^\top D X_{\text{cyc}}\big)^{-1} X_{\text{cyc}}^\top D p$ (feature disentanglement).
6. Interpret the final p as a *policy-identifiable preference over cycles*, orthogonal to the engineered features.

14 Response 1

14.1 Cycle/potential projector: algebra check

14.1.1 Dimensions

- $B \in \mathbb{R}^{(n-1)\times m}$, $W \in \mathbb{R}^{m\times m}$ diagonal with $W \succ 0$.
- $v \in \mathbb{R}^m$, $\phi \in \mathbb{R}^{n-1}$.
- $BWv \in \mathbb{R}^{n-1}$, $L := BWB^\top \in \mathbb{R}^{(n-1)\times(n-1)}$.

All products in

$$P_{\text{cyc}} = I - B^\top(BWB^\top)^{-1}BW \tag{2}$$

have consistent shape: $B^\top(\cdot)BW$ is $m \times m$.

14.1.2 Weighted least-squares → normal equations

The objective is

$$\min_{\phi} \,(v - B^\top \phi)^\top W \,(v - B^\top \phi). \tag{3}$$

The gradient w.r.t. ϕ is

$$\nabla_\phi = -2\,B\,W\,(v - B^\top \phi), \tag{4}$$

so first-order optimality gives

$$BW(v - B^\top \phi^\star) = 0 \iff (BWB^\top)\phi^\star = BWv. \tag{5}$$

Hence

$$\phi^\star = (BWB^\top)^{-1} BWv, \qquad P_{\text{cyc}} v = v - B^\top \phi^\star. \tag{6}$$

These expressions are correct.

14.1.3 Projector properties (quick sanity)

Let $L := BWB^\top$. The following identities hold.

1) $BW(P_{\mathbf{cyc}} v) = 0$

$$\begin{aligned} BW\big(I - B^\top L^{-1} BW\big)v &= BWv - (BWB^\top)L^{-1}BWv \qquad (7)\\ &= BWv - BWv = 0. \qquad (8) \end{aligned}$$

2) $P_{\mathbf{cyc}}(v + B^\top \psi) = P_{\mathbf{cyc}} v$ (because $P_{\text{cyc}} B^\top = 0$):

$$\begin{aligned} \big(I - B^\top L^{-1} BW\big) B^\top \psi &= B^\top \psi - B^\top L^{-1}(BWB^\top)\psi \qquad (9)\\ &= B^\top \psi - B^\top \psi = 0. \qquad (10) \end{aligned}$$

3) **Idempotence:** $P_{\mathbf{cyc}}^2 = P_{\mathbf{cyc}}$ Using $BWB^\top = L$ one verifies P_{cyc} is idempotent.

Thus P_{cyc} removes the potential component provided L is invertible (reduced/anchored case, connected graph, $W \succ 0$).

14.2 Weighted cycle space definition

The authors define

$$\mathcal{C}_W := \ker(BW). \tag{11}$$

Since

$$\langle B^\top \phi,\, z\rangle_W = (B^\top \phi)^\top W z = \phi^\top BWz,$$

$\mathcal{C}_W$ is exactly the W-orthogonal complement of $\operatorname{im}(B^\top)$. Because W is invertible, $\ker(BW) = W^{-1}\ker(B)$, i.e. a scaled version of the usual cycle space.

14.3 Pre-projecting X: algebra check

Define $X_{\text{cyc}} := P_{\text{cyc}}X$. Then

$$BWX_{\text{cyc}} = BW\big(I - B^\top L^{-1}BW\big)X \tag{12}$$
$$= BWX - (BWB^\top)L^{-1}BWX \tag{13}$$
$$= BWX - BWX = 0. \tag{14}$$

Hence every column of X_{cyc} lies in $\ker(BW)$, and for any α

$$BW\big(X_{\text{cyc}}\alpha\big) = 0. \tag{15}$$

Thus if $p \in \ker(BW)$ then the update $p \leftarrow p - X_{\text{cyc}}\alpha$ stays in $\ker(BW)$.

14.4 Feature-orthogonal projection step

The authors enforce $X_{\text{cyc}}^\top Dp = 0$ by

$$p \leftarrow p - X_{\text{cyc}}\alpha, \qquad \big(X_{\text{cyc}}^\top DX_{\text{cyc}}\big)\alpha = X_{\text{cyc}}^\top Dp. \tag{16}$$

This is the standard D-orthogonal projection onto $\ker(X_{\text{cyc}}^\top D)$ along $\operatorname{span}(X_{\text{cyc}})$, assuming $X_{\text{cyc}}^\top DX_{\text{cyc}}$ is invertible on that subspace.

If a ridge term $G := X_{\text{cyc}}^\top DX_{\text{cyc}} + \gamma I$ is used, solving

$$G\alpha = X_{\text{cyc}}^\top Dp$$

yields

$$X_{\text{cyc}}^\top D\big(p - X_{\text{cyc}}\alpha\big) = \gamma\alpha,$$

so the constraint $X_{\text{cyc}}^\top Dp = 0$ holds exactly only when $\gamma = 0$. With $\gamma > 0$ the step provides a *regularized approximate* projection.

14.5 Two-step projection onto the intersection

The two-step procedure reaches a point in the intersection because:

1) Step 1 maps to $\ker(BW)$.

2) Step 2 subtracts a vector in $\operatorname{span}(X_{\text{cyc}}) \subseteq \ker(BW)$, preserving membership in $\ker(BW)$, and is constructed to enforce $X_{\text{cyc}}^\top Dp = 0$.

Thus feasibility in one pass is correct. The resulting operator is a linear projector onto the intersection, but not necessarily the *orthogonal* projector for a single global metric unless that metric is chosen appropriately.

14.6 Clarifying the $BW\phi = 0$ confusion

1) The equation $BW\phi = 0$ is dimensionally inconsistent: BW maps edge vectors $\in \mathbb{R}^m$ to node-balance vectors $\in \mathbb{R}^{n-1}$, whereas $\phi \in \mathbb{R}^{n-1}$.

2) One usually does *not* constrain ϕ at all. Instead, fix a gauge on the edge-cost vector by applying P_{cyc}.

A clean formulation is

$$\tilde{c} := P_{\text{cyc}} c, \qquad \tilde{c} = X_{\text{cyc}} w + p,\ p \in \ker(BW),\ X_{\text{cyc}}^\top D p = 0.$$

Then ϕ disappears automatically after projection, and no extra coupling constraint between p and ϕ is required.

14.7 Bottom line

- The projector $P_{\text{cyc}} = I - B^\top (BWB^\top)^{-1} BW$ and the associated algebra are correct.
- Pre-projecting X indeed makes the two constraints compatible, so the two-step "make p cyclic, then make it feature-orthogonal" yields a feasible p in one pass.
- Avoid writing $BW\phi = 0$; the gauge should be imposed on c or p, not on ϕ.
- Adding a ridge term destroys the exactness of the $X_{\text{cyc}}^\top D p = 0$ constraint; set $\gamma = 0$ for exact feasibility or accept a regularized approximation.
- The two-step method is a valid linear projection onto the intersection, but it is not necessarily the orthogonal projection for a single quadratic metric.

15 Response 2

15.1 Cheapest monitor: cycle-feasibility residual

For any edge vector v (e.g., the current p or a temporary update p_{tmp}), define

$$r_{\text{cyc}}(v) := \|BWv\|_2. \tag{17}$$

- $r_{\text{cyc}}(v) = 0$ iff $v \in \ker(BW)$.
- In practice one monitors this quantity *before* applying the cycle projection, because after projection it will be (up to round-off) zero.

15.2 Potential energy via a Laplacian solve

Let $L := BWB^\top$ and compute the potential fit

$$\phi^\star(v) := L^{-1}BWv. \tag{18}$$

The removed potential component is $g(v) = B^\top \phi^\star(v)$, whose W-energy is

$$\begin{aligned} E_{\mathrm{pot}}(v) &:= \|g(v)\|_W^2 && (19)\\ &= \|B^\top \phi^\star(v)\|_W^2 && (20)\\ &= \phi^\star(v)^\top L\, \phi^\star(v) && (21)\\ &= (BWv)^\top L^{-1}(BWv). && (22) \end{aligned}$$

Since P_{cyc} is the W-orthogonal projector, the energy splits exactly:

$$\|v\|_W^2 = \|P_{\mathrm{cyc}} v\|_W^2 + E_{\mathrm{pot}}(v). \tag{23}$$

A convenient normalized "cycle fraction" is

$$s_{\mathrm{cyc}}(v) := \frac{\|P_{\mathrm{cyc}} v\|_W^2}{\|v\|_W^2} = 1 - \frac{(BWv)^\top L^{-1}(BWv)}{v^\top W v}. \tag{24}$$

15.3 Graph/weight "health" monitor: conditioning of L

The numerical stability of the previous quantities is governed by the spectrum of $L = BWB^\top$.

- $\lambda_{\min}(L)$: a very small value indicates an ill-conditioned Laplacian solve.
- $\kappa(L) = \lambda_{\max}(L)/\lambda_{\min}(L)$: a large condition number signals potential sensitivity in the potential estimates and projections.

15.4 Number of cyclic degrees of freedom

The dimension of the weighted cycle space is

$$\dim \ker(BW) = m - \operatorname{rank}(BW). \tag{25}$$

For a connected graph with one anchored node and $W \succ 0$, $\operatorname{rank}(BW) = n - 1$, so

$$\dim \ker(BW) = m - n + 1. \tag{26}$$

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